Matrix Trader 7.0

YouTube Strategy Testing Research Compilation

How the 100,000+ strategy and 350,000 Bitcoin strategy tests can improve MT7 edge, validation, and profitability

Core thesis

The most valuable idea from both videos is not a magic indicator. It is the validation system: walk-forward testing, out-of-sample survival, regime awareness, robustness filters, and forcing most strategies to fail before they ever reach live signal generation.

Generated May 31, 2026

1. Executive Summary

This PDF compiles the full Matrix Trader conversation around two YouTube strategy-testing videos, the screenshots provided, and the two transcript files uploaded by the user.

The first video tested a broad universe of retail strategies across multiple assets. The second narrowed the question to Bitcoin across multiple timeframes. Together, they point to a major design principle for MT7: strategy selection should be regime-dependent, asset-dependent, and validated out-of-sample before it is trusted.

MT7 takeaway

Do not simply import the winning strategies from the videos. Use the methodology to build a crypto-native Edge Lab that tests MEXC perpetual futures strategies with funding, open interest, liquidation zones, VWAP, premium/basis, depth, spread, and fees/slippage.

High-Level Conclusions

- The first video is strongest as a validation-framework reference: 131,441 backtests, 62 strategies, 31 assets, and only 65 final survivors after walk-forward and filters.
- The second video is more directly relevant to MT7 because it focuses on Bitcoin, multiple intraday timeframes, native exchange data, and regime instability.
- For BTC, the surviving logic shifted toward trend-following, volatility breakout, and adaptive trend/reversion systems rather than pure mean reversion.
- For high-leverage MEXC perps, drawdown tolerance from the video cannot be copied directly. MT7 needs ATR stops, leverage throttles, spread filters, daily loss caps, and kill switches.
- The best next step is not one new strategy, but a Verified Strategy Registry backed by walk-forward validation and a Regime Router.

2. Conversation Record - What Was Discussed

Initial user question

The user asked how useful the concepts, topics, and strategies from a YouTube video could be for Matrix Trader, with the goal of testing the best-performing concepts to improve MT7 edge and profitability. The user shared screenshots and later uploaded the transcript for the video titled "I Tested 100,000 Trading Strategies."

Initial assistant assessment

The first answer concluded that the video was valuable not mainly because of any single indicator, but because of the systematic validation framework: in-sample testing, out-of-sample testing, walk-forward verification, overfitting detection, robustness filters, ranking, and quality scoring.

The initial recommendation was to build MT7 around a Strategy Laboratory, Verification Engine, Robustness Score, and Verified Strategy Registry rather than simply copying retail indicators.

Transcript-based update

After the user uploaded the transcript, the analysis became more confident. The transcript clarified that the video was fundamentally about overfitting, survivorship bias, walk-forward testing, and the importance of out-of-sample survival. The recommendation became: do not blindly implement the top video strategies. Instead, use the framework to discover crypto-native strategies.

Second video update

The user then provided an additional video, screenshots, and transcript titled "I Tested 350,000 Bitcoin Strategies. 4 Survived." The analysis updated again: Bitcoin behaved differently from the broad equity/ETF test. Mean reversion dominated the broad test, but BTC survivors were mostly trend-following, volatility breakout, and adaptive systems.

Final user request

The user asked to compile the entire conversation into a PDF for sharing and later reference.

3. Video 1 - 100,000+ Strategy Test

Source material: uploaded transcript titled "I Tested 100,000 Trading Strategies..txt" plus screenshots of the notebook/results interface.

What the video tested

- 131,441 individual backtest runs.
- 62 strategies.
- 31 different assets.
- Up to 100 Bayesian-optimized parameter combinations per strategy per asset.
- 917 results looked promising in-sample.
- 358 survived walk-forward validation.
- 65 survived the final six filters.

Core methodology

- Optimize on the first 70% of each sequential window.
- Test completely blind on the remaining 30% of each window.
- Move the window forward and repeat.
- Stitch all out-of-sample windows into one combined OOS result.
- Treat the stitched OOS result as the only number that matters.

Six filters applied after walk-forward testing

Filter	Meaning for MT7
OOS drawdown better than -35%	Reject strategies requiring unrealistic pain tolerance.
OOS Sharpe above 0.5	Reject performance statistically close to noise.
OOS Sharpe below 2.5	Reject suspiciously perfect results that may be artifact/asset drift.
OOS Sharpe cannot exceed IS Sharpe by more than 30%	Reject lucky out-of-sample variance masquerading as edge.
Minimum 30 trades	Reject samples too small to trust.
Positive in-sample Sharpe	Basic sanity check before considering OOS behavior.

Important findings

- Mean reversion dominated the verified set across the broad asset universe.
- Volume strategies and mean reversion had the strongest category survival rates in the broad test.
- Trend-following and candlestick pattern strategies had weaker survival rates across the broad multi-asset sample.
- Simple strategies with fewer parameters survived better because they leave less room for curve fitting.
- Strategies that looked spectacular in-sample often collapsed out-of-sample.

Initial MT7 implication

For MT7, the first video supports building a systematic Edge Lab. The goal is not to ask "what strategy looked best?" but rather "what strategy survives unseen markets, multiple regimes, realistic frictions, and crypto-native conditions?"

```
MT7 Edge Lab principle:

Generate many strategy candidates
-> walk-forward test them
-> apply robustness filters
-> classify by regime
```

```
  -> only promote survivors to Verified Strategy Registry
```

4. Video 2 - Bitcoin Multi-Timeframe Test

Source material: uploaded transcript titled "I Tested 350,000 Bitcoin Strategies. 4 Survived..txt" plus screenshots of the Bitcoin notebook and pipeline.

What the Bitcoin test changed

The second video corrected the broad takeaway from the first. Across equities and ETFs, mean reversion survived strongly. On Bitcoin, the standard retail indicator toolkit performed much worse. The video says the broad study found 65 survivors, but on Bitcoin the result was zero under the stricter drawdown threshold. This is crucial for MT7 because it proves that strategy logic must be asset-specific and regime-specific.

Data and pipeline details

- BTCUSDT tested across 5m, 15m, 1h, 4h, and 1d timeframes.
- Native Binance data from January 2019 onward, not simply resampled data.
- More than 1 million individual price bars before strategy testing.
- 70 strategies tested through Bayesian optimization.
- Parameter ranges scaled per timeframe so lower-timeframe lookbacks were not naively copied from daily settings.
- Initial full run took roughly 4 hours and 42 minutes; cached rerun finished in roughly 6.5 minutes.

BTC survival funnel

Stage	Result
Generated/tested results with enough trades	17,856
Walk-forward survivors	18
Survivors with max drawdown -40%	0
Survivors with max drawdown -50%	2
Survivors with max drawdown -60%	4

Critical warning for high leverage

A strategy that needs a -60% equity-curve drawdown tolerance on spot-style testing cannot be copied directly into 20x-50x perp trading. MT7 must convert any raw strategy into a controlled execution model with small per-trade risk, hard stops, spread filters, depth checks, and volatility-based leverage limits.

The four BTC survivors

Strategy	Timeframe	Why it matters for MT7
Volatility breakout	1h	Most actionable candidate for BTC trend expansion and high-volatility move detection.
Parabolic SAR	1d	Useful as a higher-timeframe trend regime filter, not necessarily an entry trigger.
Dual momentum	4h	Useful directional confirmation before lower-timeframe execution.
Mean-reversion/trend adaptive	4h	Useful because it changes behavior depending on measured market conditions.

Most important Bitcoin-specific insight

The second video says Bitcoin is not naturally friendly to pure mean reversion across the tested period. It went through structurally different regimes: bear market, explosive bull run, major crash, and a different recovery cycle.

Each regime had different volatility, trend persistence, and mean-reversion behavior. Walk-forward testing exposes this because a strategy cannot hide inside the one period where it worked.

```
BTC-specific rule for MT7:

Do not fade BTC by default.
First detect trend regime and volatility expansion.
Only consider reversion when OI, funding, liquidation exhaustion, and VWAP stretch all support
exhaustion.
```

5. Combined Meta-Analysis For Matrix Trader 7.0

The two videos are not contradictory

The first video says mean reversion survived best across a broad equity/ETF/asset universe. The second says Bitcoin needs trend-following, volatility breakout, and adaptive logic. The combined lesson is that there is no universal strategy. MT7 should route strategies by asset, timeframe, volatility regime, funding regime, and liquidity structure.

What MT7 should take directly

- Walk-forward testing as a non-negotiable part of strategy promotion.
- Out-of-sample Sharpe, OOS return, OOS drawdown, and trade count for every strategy.
- Overfitting diagnostics comparing in-sample and out-of-sample behavior.
- A robustness/quality score that penalizes low trade count, high drawdown, and unstable results.
- A database-backed research pipeline that stores every result and skips already-computed combinations.
- A strict distinction between research-only strategies and live-approved strategies.

What MT7 should not copy blindly

- Do not assume Binance spot results transfer directly to MEXC perpetual swaps.
- Do not assume BTC behavior transfers to smaller altcoin perps.
- Do not use daily-bar drawdown thresholds as safe thresholds for high-leverage intraday trading.
- Do not treat RSI, Bollinger Bands, Parabolic SAR, or any indicator as alpha by itself.
- Do not let a strategy go live just because it has strong raw return or one attractive equity curve.

Where MT7 can surpass the videos

The second video explicitly did not test order flow, funding rates, on-chain signals, or portfolio combinations. Those are exactly the areas where MT7 has the opportunity to become stronger than a retail-indicator testing framework.

Video limitation	MT7 upgrade path
No funding rates	Add funding extremes, funding z-score, funding direction changes, and funding/price divergence.
No open interest	Add OI expansion/contraction, OI divergence, and OI flush logic.
No liquidation data	Add liquidation cascade and exhaustion modules.
No perp premium/basis	Add premium distortion and basis normalization.
No depth/spread model	Add executable liquidity checks and slippage estimates.
No portfolio combinations	Test strategy baskets rather than one strategy at a time.

6. Proposed MT7 Regime Router

The videos strengthen the case for a Regime Router. Before any strategy fires, MT7 should classify the current market state and only allow strategies that historically survive in that regime.

Regime	Definition	Allowed strategy families
TRENDING_UP	Higher highs/higher lows, price above VWAP, momentum aligned.	Volatility breakout, dual momentum, trend continuation.

TRENDING_DOWN	Lower highs/lower lows, price below VWAP, sell pressure persistent.	Short continuation, downside breakout, liquidation cascade follow-through.
VOL_EXPANSION	ATR/realized volatility expanding from compression.	Breakout and momentum systems.
VOL_COMPRESSION	ATR/realized volatility contracting; range becoming tight.	Squeeze watchlist only until breakout confirmation.
RANGE_BOUND	Price oscillates around VWAP with low directional persistence.	VWAP reversion, percent-B reversion, small-risk mean reversion.
FUNDING_DISTORTION	Funding extreme relative to its own history.	Funding mean reversion or contrarian setup with confirmation.
LIQUIDATION_CASCADE	Large forced liquidations and fast displacement.	Cascade follow-through or exhaustion reversion, depending on OI/funding/depth.
OI_EXPANSION	Open interest rising with price movement.	Trend continuation if direction and liquidity support it.
OI_FLUSH	Open interest collapsing after a violent move.	Exhaustion/reversion candidate if price stabilizes.

Design principle

Strategy selection should be regime-dependent, not indicator-dependent. MT7 should not ask whether RSI is good. It should ask whether this strategy family has survived this specific asset, timeframe, liquidity condition, funding condition, and volatility regime.

7. Strategy Candidates To Test Next

Tier 1: BTC/Major-Cap Trend Engine

This is the cleanest bridge between the second video and MT7.

```
BTC 1h Volatility Breakout
+ 4h Dual Momentum Confirmation
+ 1d Parabolic SAR Regime Filter
+ MEXC OI/Funding/Liquidation Confirmation
+ ATR-Based Leverage Throttle
```

- Primary market: BTCUSDT and ETHUSDT perps first, then SOL and other liquid majors.
- Entry should require volatility expansion, price displacement, and directional confirmation.
- Execution should be blocked if spread, depth, ADL, or funding distortion makes the trade unsafe.
- Use 1h for breakout detection, 4h for directional confirmation, and 1d for macro trend filter.

Tier 1: Crypto-Native Mean Reversion Lab

The first video supports mean reversion broadly, but the second warns against pure BTC fading. For MT7, mean reversion should be crypto-native and perp-aware.

```
Funding extreme
+ OI flush or liquidation exhaustion
+ VWAP/ATR stretch
+ depth imbalance improvement
+ price stabilization
= possible reversion setup
```

- Do not trade simple RSI oversold/overbought alone.
- Require funding, OI, liquidation, VWAP, and liquidity context.
- Treat reversion as a setup family, not as a single indicator.

Tier 2: Portfolio Strategy Basket

The second video notes that combinations of multiple strategies often improve risk-adjusted performance compared with single strategies. MT7 should test baskets.

Basket component	Purpose
BTC 1h volatility breakout	Capture major expansion phases.
BTC/ETH 4h dual momentum	Confirm direction and reduce false breakouts.
Altcoin funding squeeze	Exploit perp-specific positioning distortions.
Liquidation exhaustion reversion	Capture forced-selling/forced-buying reversals.
Market-wide risk-off filter	Prevent trades during hostile macro/liquidity conditions.

8. Recommended Implementation Plan

1. Create a research-only Edge Lab namespace for new strategies. Nothing new should be live-approved by default.
2. Implement the four BTC survivors as baseline candidates: volatility_breakout, dual_momentum, parabolic_sar_trend, and mean_rev_trend_adaptive.
3. Retest them on MEXC perpetual futures, not Binance spot, using candles plus fees, spread, slippage, funding, OI, and liquidation data where available.
4. Build a Regime Router and attach every backtest result to regime labels so MT7 learns where each strategy actually works.
5. Create a Verified Strategy Registry with statuses: RESEARCH_ONLY, WATCHLIST, VERIFIED, LIVE_APPROVED, RETIRED, FAILED.
6. Add portfolio combination testing so MT7 can evaluate strategy baskets rather than overfitting to one isolated indicator system.
7. Add high-leverage risk gates before any strategy can trade: ATR stop, max trade risk, leverage cap, spread/depth check, daily loss cap, consecutive-loss throttle, and kill switch.

Suggested registry schema

```
strategy_id
symbol
timeframe
regime_tags
status
is_sharpe
oos_sharpe
oos_return_pct
oos_max_drawdown_pct
trade_count
profit_factor
win_rate
fees_model
slippage_model
funding_context
oi_context
liq_context
quality_score
last_validated_at
```

Promotion criteria

Criterion	Suggested MT7 rule
Trade count	Minimum 100+ for intraday strategies when possible; 30 is a floor, not ideal.
OOS Sharpe	Positive and stable across walk-forward windows.
Drawdown	Acceptable only after translating to per-trade risk and leverage-safe execution.
Regime survival	Must identify at least one regime where the strategy survives consistently.
Fee/slippage robustness	Must remain positive after realistic MEXC perp execution costs.
No single-window dependency	Cannot rely on one lucky bull/bear period.
Live approval	Only after paper/shadow mode agrees with backtest

	behavior.

9. Blind Spots, Risks, And Caveats

Major caveats from the videos

- The broad video was mostly equity/ETF-centric, so its strategy rankings do not transfer cleanly to crypto perps.
- The Bitcoin video tested standard retail indicators and did not include the most important crypto-native data sources.
- The Bitcoin video used Binance data, not MEXC perps. Exchange microstructure differences matter.
- A drawdown threshold of -60% is unacceptable for high-leverage execution unless risk is transformed at the trade level.
- Bayesian optimization can still overfit if the validation design, constraints, or objective function are poorly chosen.
- Walk-forward testing is necessary but not sufficient. MT7 also needs live shadow testing and ongoing degradation monitoring.

MT7-specific risks

- Too many indicators can create a sophisticated curve fit that looks intelligent but fails live.
- Extreme volatility is not always opportunity. Your prior MT7 audit showed extreme-vol regimes were deeply negative overall.
- Data quality problems, missing candles, stale API values, or exchange-specific restrictions can create false confidence.
- High leverage magnifies small assumptions about slippage, spread, and stop execution into account-level risk.
- Strategy baskets can improve robustness, but they can also hide correlated failure if all strategies depend on the same market regime.

Survival rule

The goal is not to maximize backtest return. The goal is to maximize the odds that MT7 survives long enough to compound real edge. Every new strategy must be evaluated through that lens.

10. Final MT7 Blueprint From This Conversation

The best near-term architecture is a research and validation layer that turns MT7 into a machine for finding, killing, and promoting strategies based on evidence.

```
Data Ingestion
  -> MEXC candles, funding, OI, liquidations, depth, spread, premium/basis

Feature Layer
  -> VWAP, ATR, volatility regime, OI regime, funding regime, liquidation regime

Strategy Lab
  -> generate candidates, run Bayesian optimization, store all results

Walk-Forward Engine
  -> train/test sequential windows, stitch OOS result

Regime Router
  -> classify where each strategy works and blocks it where it fails

Verified Strategy Registry
  -> research-only, watchlist, verified, live-approved, failed

Execution Gate
  -> ATR risk, leverage throttle, spread/depth, ADL/funding/premium risk, kill switch
```

Best first experiment

Start with one clear BTC experiment before expanding across all MEXC pairs:

```
Experiment: BTCUSDT Perp Trend Expansion

Signal:
- 1h volatility breakout
- 4h dual momentum confirmation
- 1d Parabolic SAR trend filter

Crypto confirmations:
- OI expansion aligned with direction
- funding not dangerously crowded
- liquidation map does not show immediate adverse magnet
- price displaced from VWAP but not excessively late

Execution controls:
- ATR stop
- risk <= 0.25%-1.0% equity per trade
- leverage capped by volatility
- no trade if spread/depth/ADL/funding gate fails
- daily loss and consecutive-loss kill switch
```

Bottom line

The videos are highly useful for Matrix Trader, but not because they reveal a magic strategy. They show what rigorous testing looks like and why most retail strategies should be rejected. MT7 should use this as a blueprint for a crypto-native, regime-aware, walk-forward validated Edge Lab. The most valuable upgrade is not another indicator. It is a system that prevents bad strategies from ever reaching live trading.

11. Optional Codex / Claude Code Handoff Prompt

This prompt can be used later as a starting point for implementation work. It is included because the conversation repeatedly pointed toward a concrete MT7 architecture update.

```
You are working inside the Matrix Trader 7.0 codebase. Implement a research-only Edge Lab upgrade
inspired by walk-forward strategy validation from recent YouTube research reviewed in our planning
notes.

Goal:
Create a crypto-native validation framework that can test candidate strategies on MEXC perpetual
futures data before any strategy is promoted to live signal generation.

Required components:
1. Strategy candidates:
   - volatility_breakout
   - dual_momentum
   - parabolic_sar_trend_filter
   - mean_rev_trend_adaptive

2. Validation pipeline:
   - sequential walk-forward train/test windows
   - stitched out-of-sample equity curve
   - OOS Sharpe, OOS return, OOS max drawdown, trade count, profit factor, win rate
   - in-sample vs out-of-sample overfitting diagnostics

3. Regime tagging:
   - trending_up, trending_down, range_bound
   - vol_expansion, vol_compression
   - funding_distortion
   - oi_expansion, oi_flush
   - liquidation_cascade, liquidation_exhaustion
```

```
4. Verified Strategy Registry:
   - RESEARCH_ONLY by default
   - WATCHLIST
   - VERIFIED
   - LIVE_APPROVED
   - FAILED
   - RETIRED

5. Safety constraints:
   - no strategy can become live-approved without realistic fees/slippage
   - no high-leverage execution without ATR stop, risk cap, spread/depth gate, and kill switch
   - Binance spot results must never be treated as MEXC perp approval

Deliverables:
- Implement modules in a clean, testable way.
- Add README documentation explaining methodology and safety constraints.
- Add example CLI commands for running BTCUSDT research-only validation.
- Do not alter live trading behavior unless explicitly requested in a later task.
```

12. Screenshot Appendix

The screenshots below were provided by the user during the conversation. They are included as visual context for the notebook outputs, tables, filters, and Bitcoin multi-timeframe pipeline discussed above.

Figure 1. Video 1 screenshot 1 - OOS Sharpe heatmap and verified strategies notebook

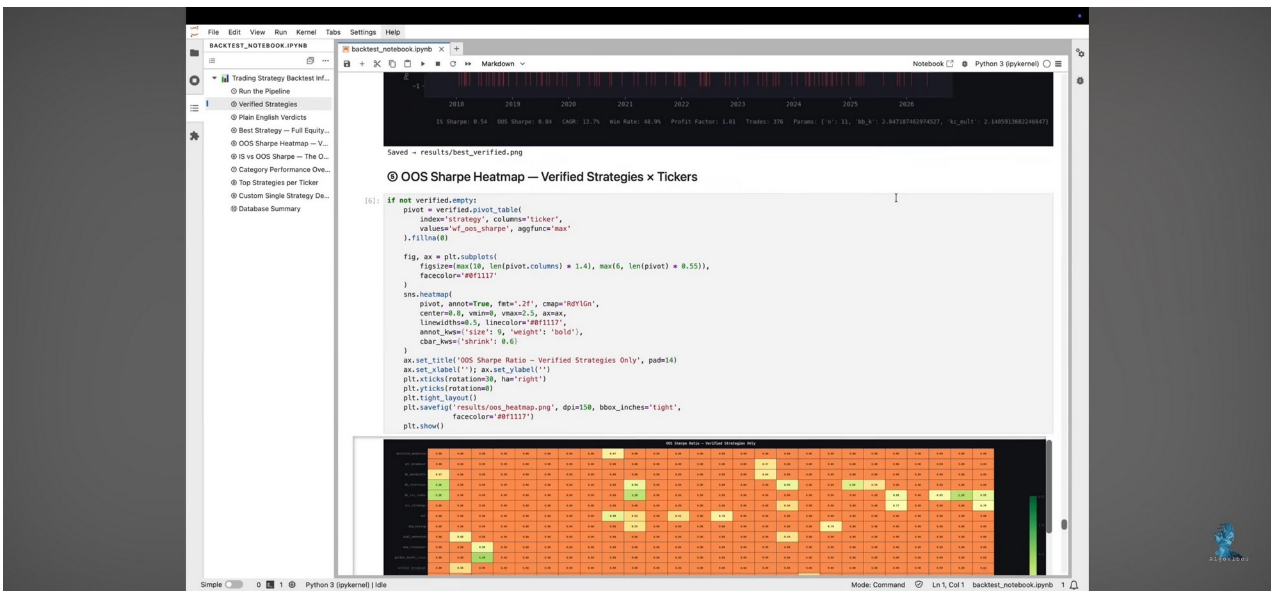


Figure 2. Video 1 screenshot 2 - Best strategy equity curve

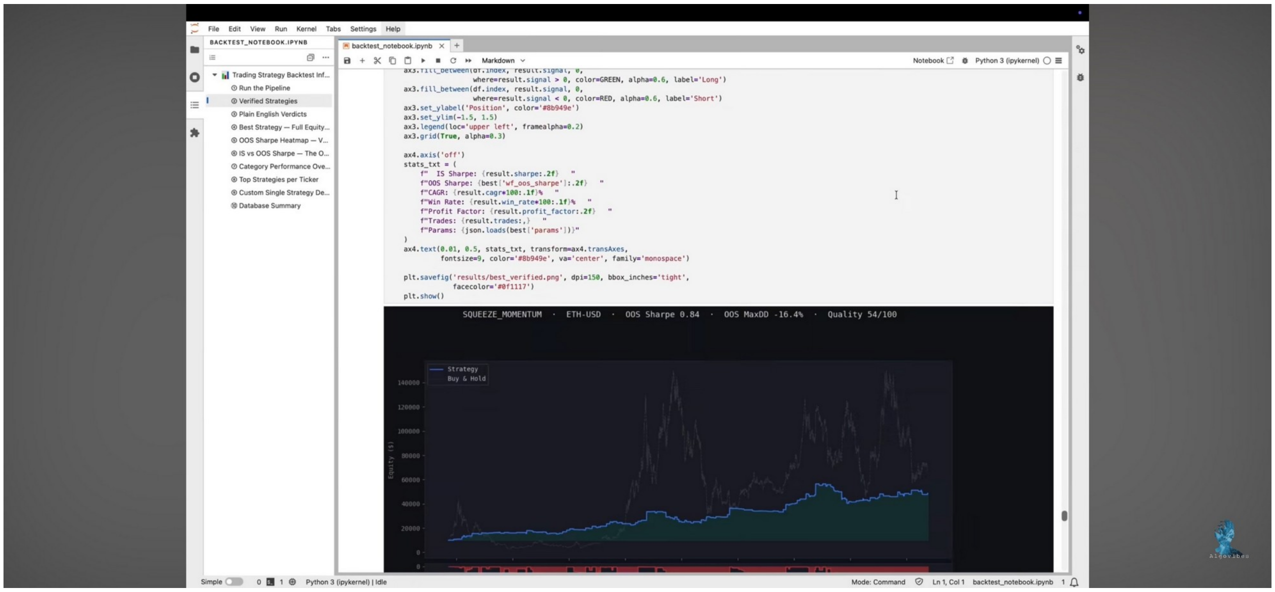


Figure 3. Video 1 screenshot 3 - Best strategy code section

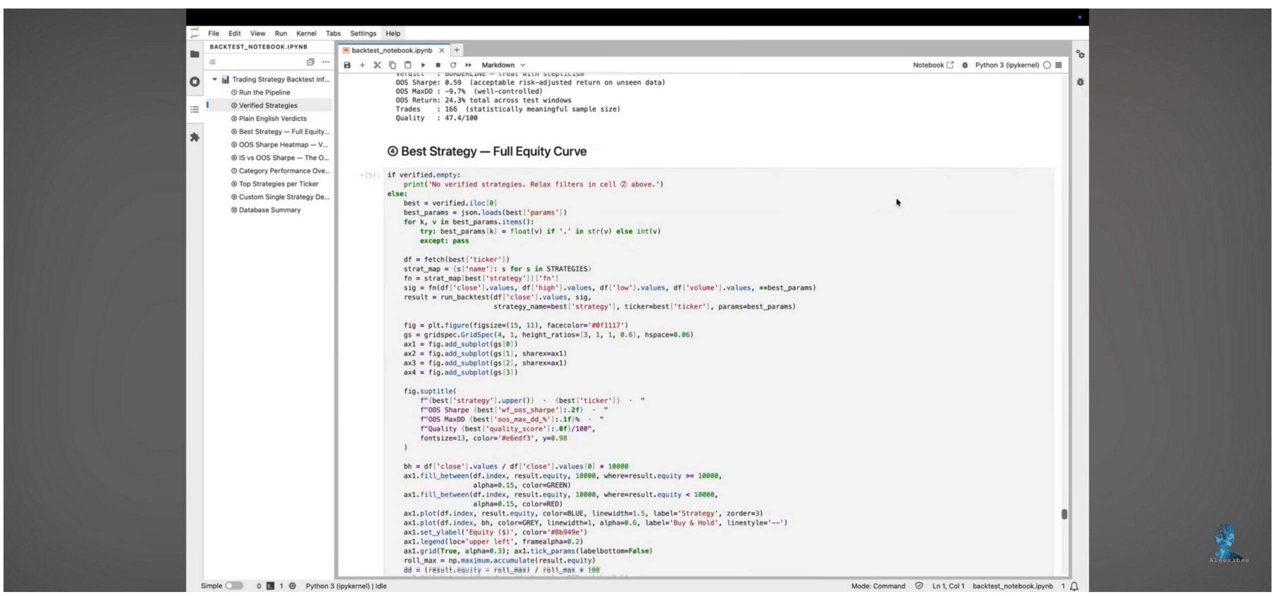


Figure 4. Video 1 screenshot 4 - Plain English verdicts and ranked table

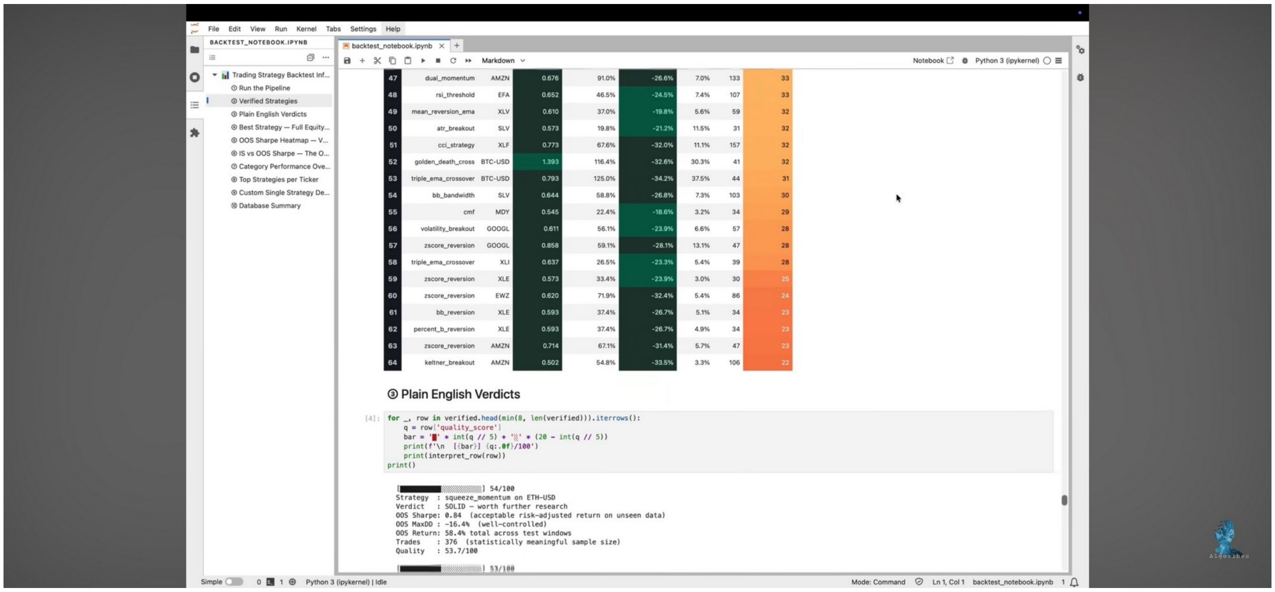


Figure 5. Video 1 screenshot 5 - Verified strategy table

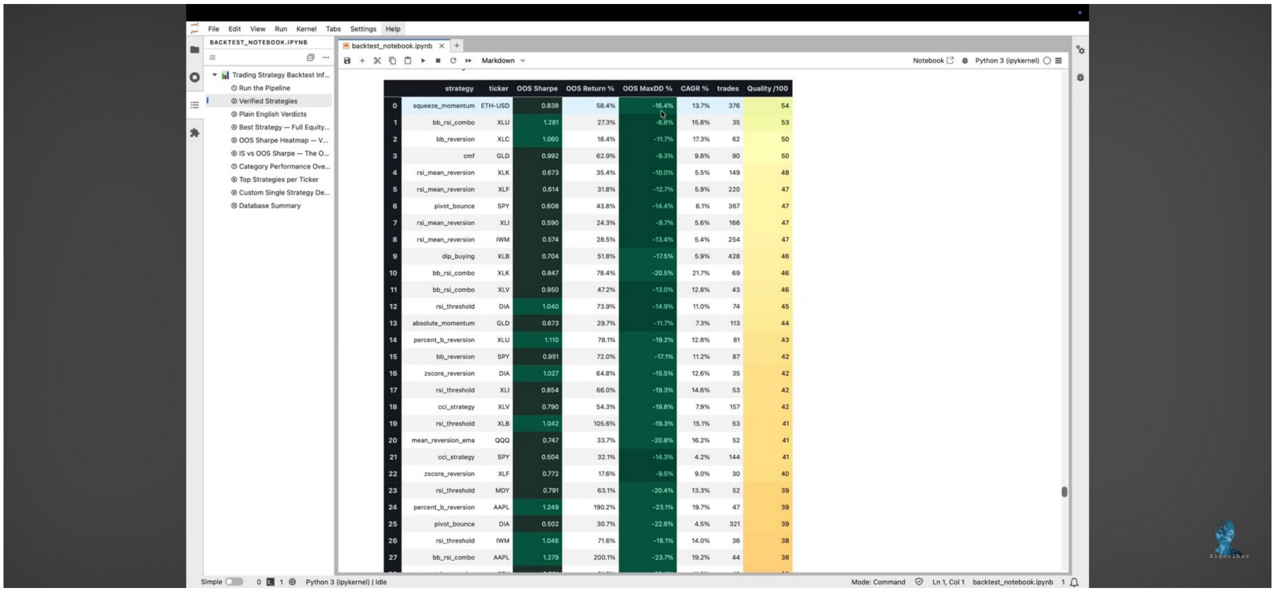


Figure 6. Video 1 screenshot 6 - Verified strategies code/filter section

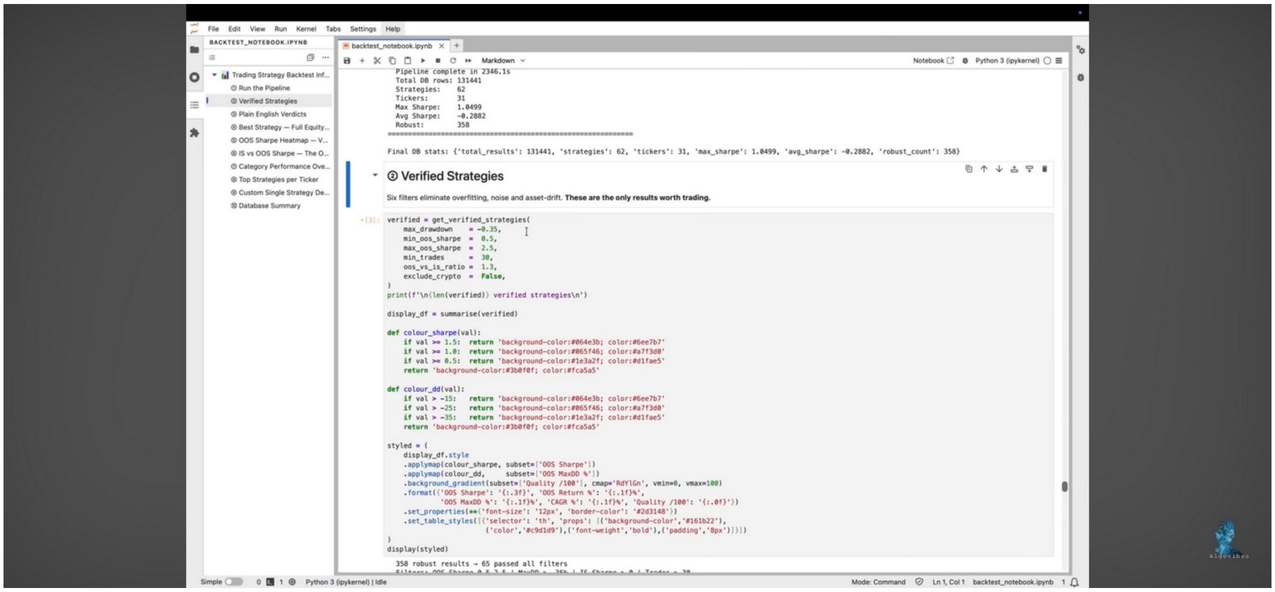


Figure 7. Video 1 screenshot 7 - IS vs OOS overfitting picture

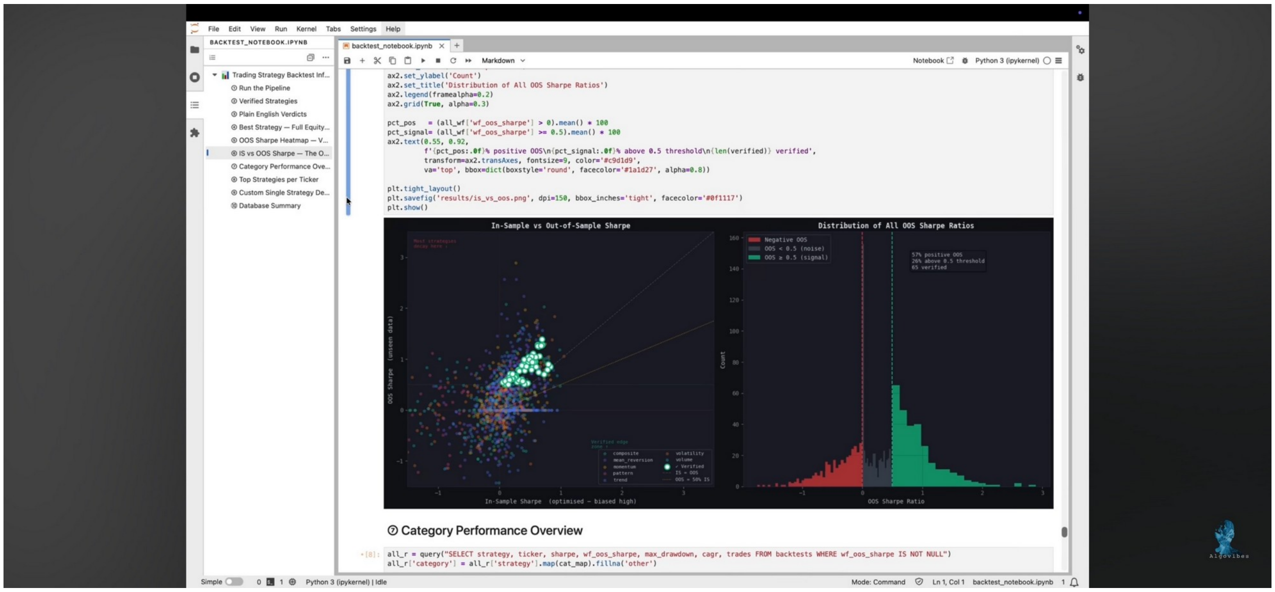


Figure 8. Video 1 screenshot 8 - Overfitting chart code

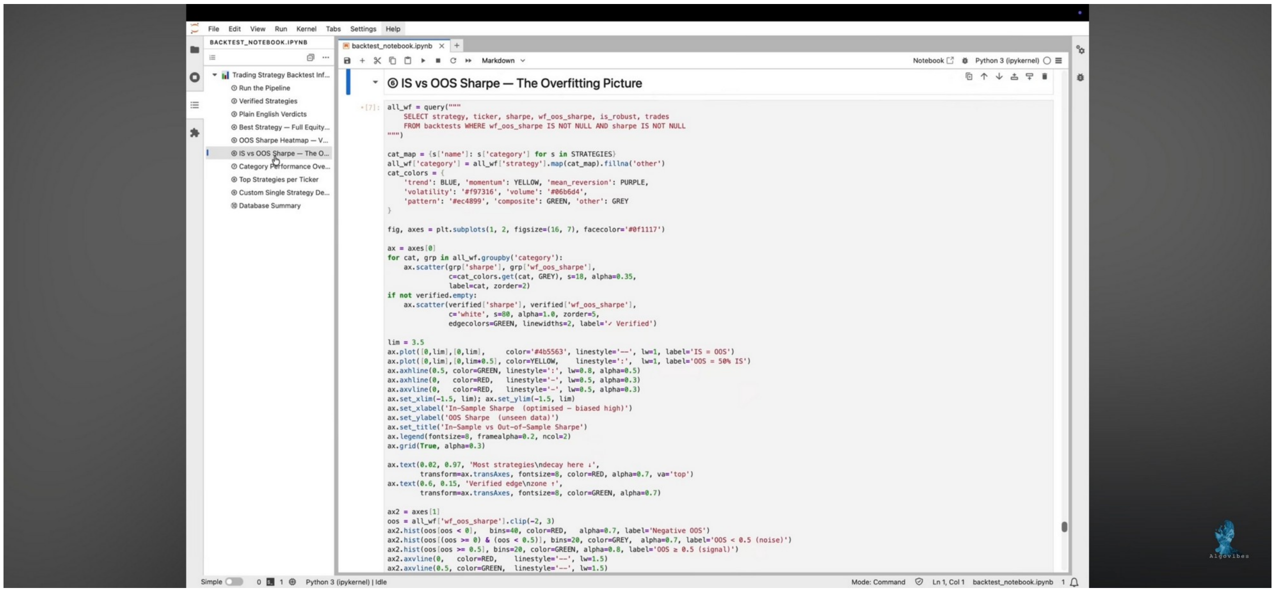


Figure 9. Video 1 screenshot 9 - Verdicts on top results

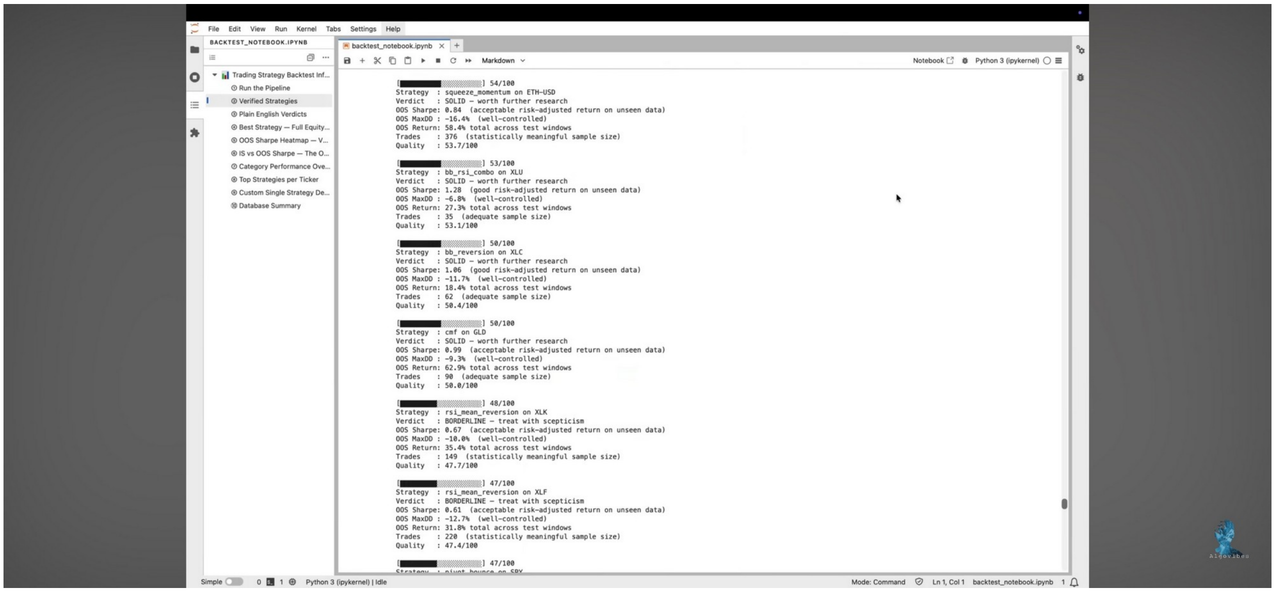


Figure 10. Video 1 screenshot 10 - Verified strategies table rerun view

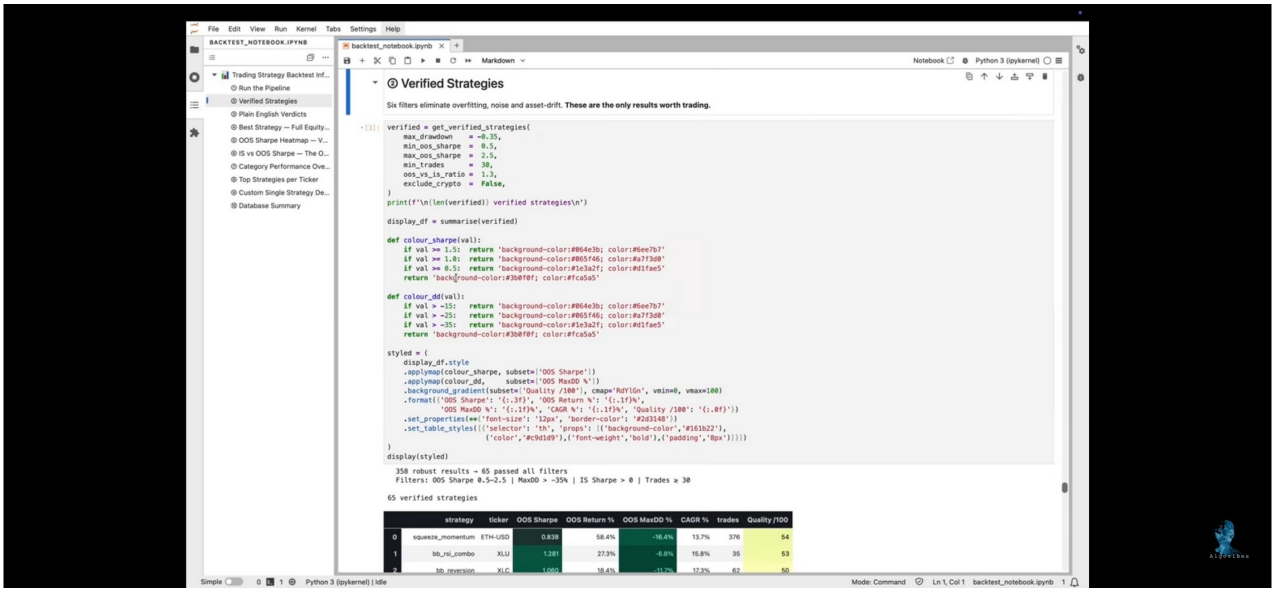


Figure 11. Video 2 screenshot 1 - Bitcoin multi-timeframe pipeline code

```python
"""
Bitcoin Multi-Timeframe Pipeline
===================================
Runs all strategies across 5m, 15m, 1h, 4h, 1d on BTCUSDT.
Target: 1,000,000+ backtest results.

KEY FEATURE: DB-skip check — already computed strategy/timeframe
combinations are skipped on re-runs. Only new work gets computed.
"""
import sys
import time
import numpy as np
from pathlib import Path

sys.path.insert(0, str(Path(__file__).parent))

from core.binance_fetcher import fetch_all_timeframes, get_cache_status
from core.backtester import BacktestConfig
from core.optimizer_tf import bayesian_search_tf, walk_forward_tf
from core.database_btc import init_db, save_batch, get_stats, deduplicate, query as btc_query
from strategies.all_strategies import STRATEGIES

TF_PARAM_OVERRIDES = {
    "5m":  {"multiplier": 0.3, "n_trials": 80},
    "15m": {"multiplier": 0.5, "n_trials": 80},
    "1h":  {"multiplier": 0.7, "n_trials": 100},
    "4h":  {"multiplier": 0.85,"n_trials": 100},
    "1d":  {"multiplier": 1.0, "n_trials": 100},
}


def scale_param_space(param_space: dict, timeframe: str) a35tet1k
    """
    Scale integer lookback windows for lower timeframes.
    Float params (multipliers, thresholds) are unchanged.
    """
    mult = TF_PARAM_OVERRIDES.get(timeframe, {}).get("multiplier", 1.0)
    if mult == 1.0:
        return param_space
    scaled = {}
    for pname, spec in param_space.items():
        if spec[0] == "int":
            lo = max(2, int(spec[1] * mult))
            hi = max(lo + 1, int(spec[2] * mult))
            scaled[pname] = ("int", lo, hi)
        else:
            scaled[pname] = spec
    return scaled
```

Figure 12. Video 2 screenshot 2 - pipeline_btc.py in JupyterLab

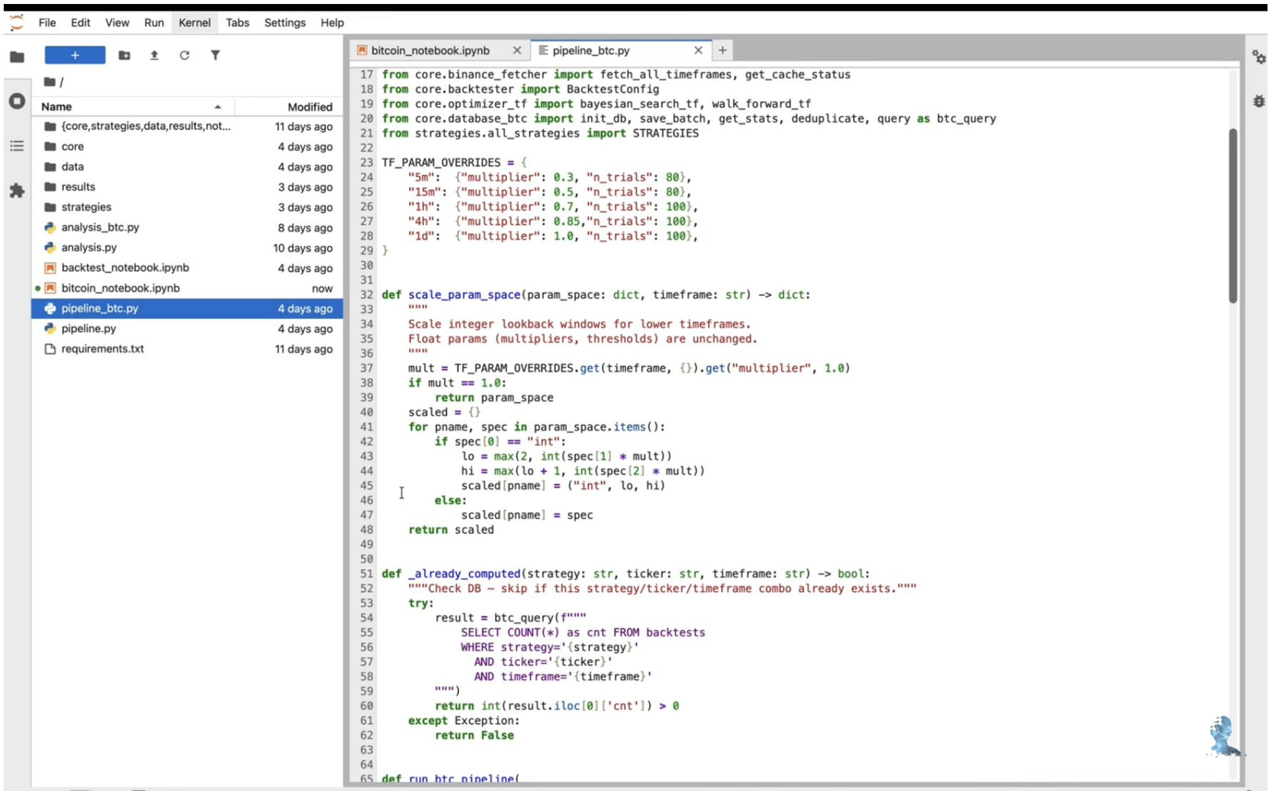


Figure 13. Video 2 screenshot 3 - Cached pipeline runtime and DB stats

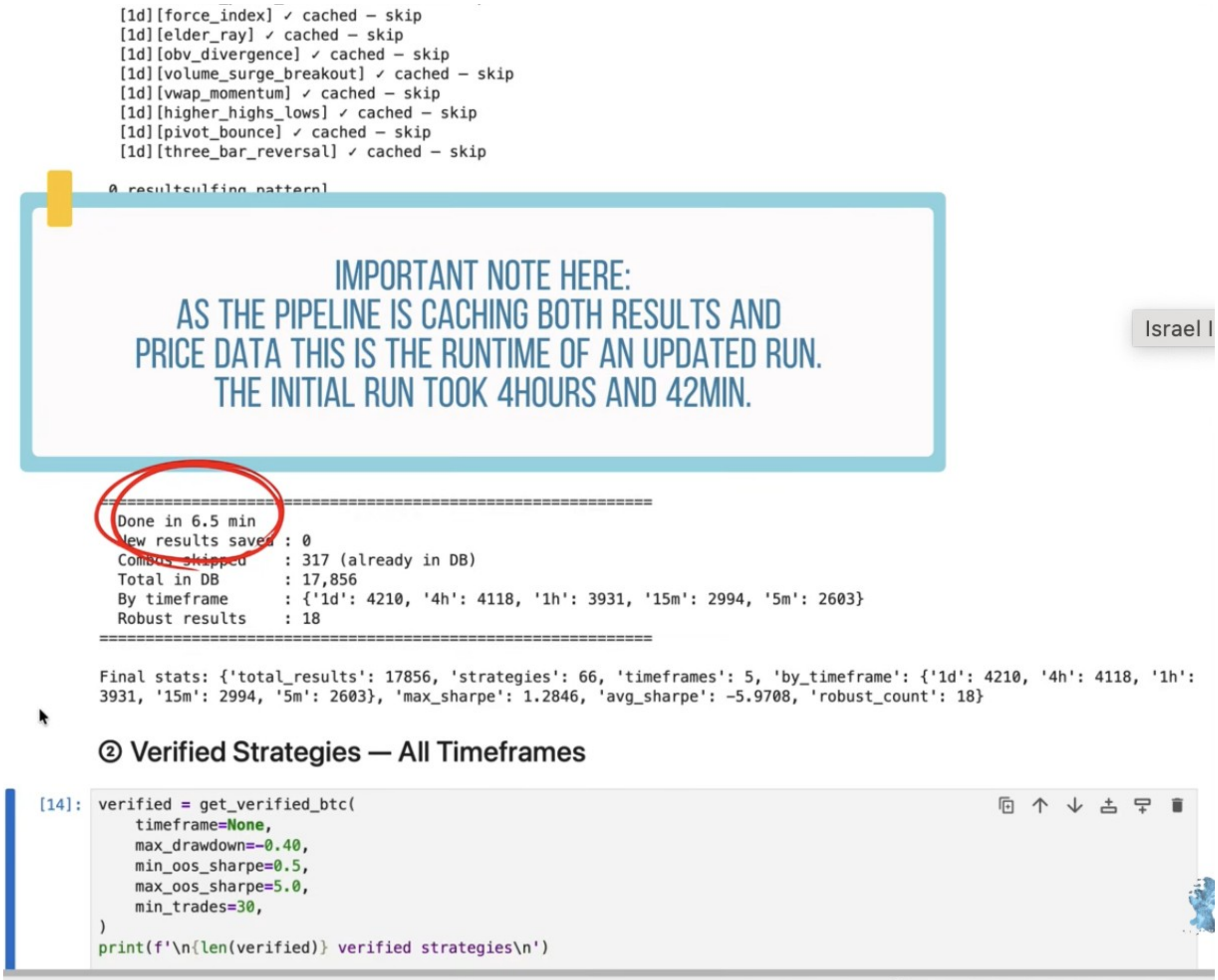


Figure 14. Video 2 screenshot 4 - Verified strategies all timeframes filters

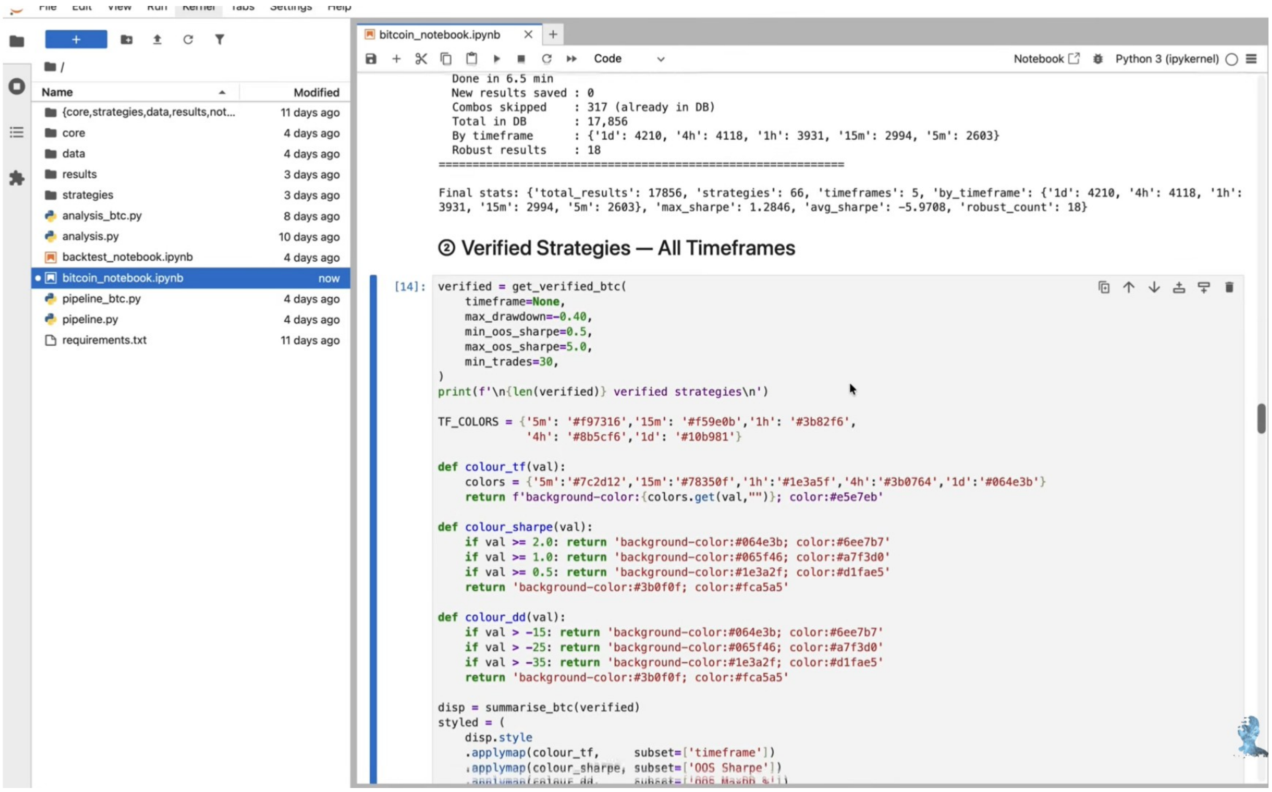


Figure 15. Video 2 screenshot 5 - Zero verified at -40% drawdown

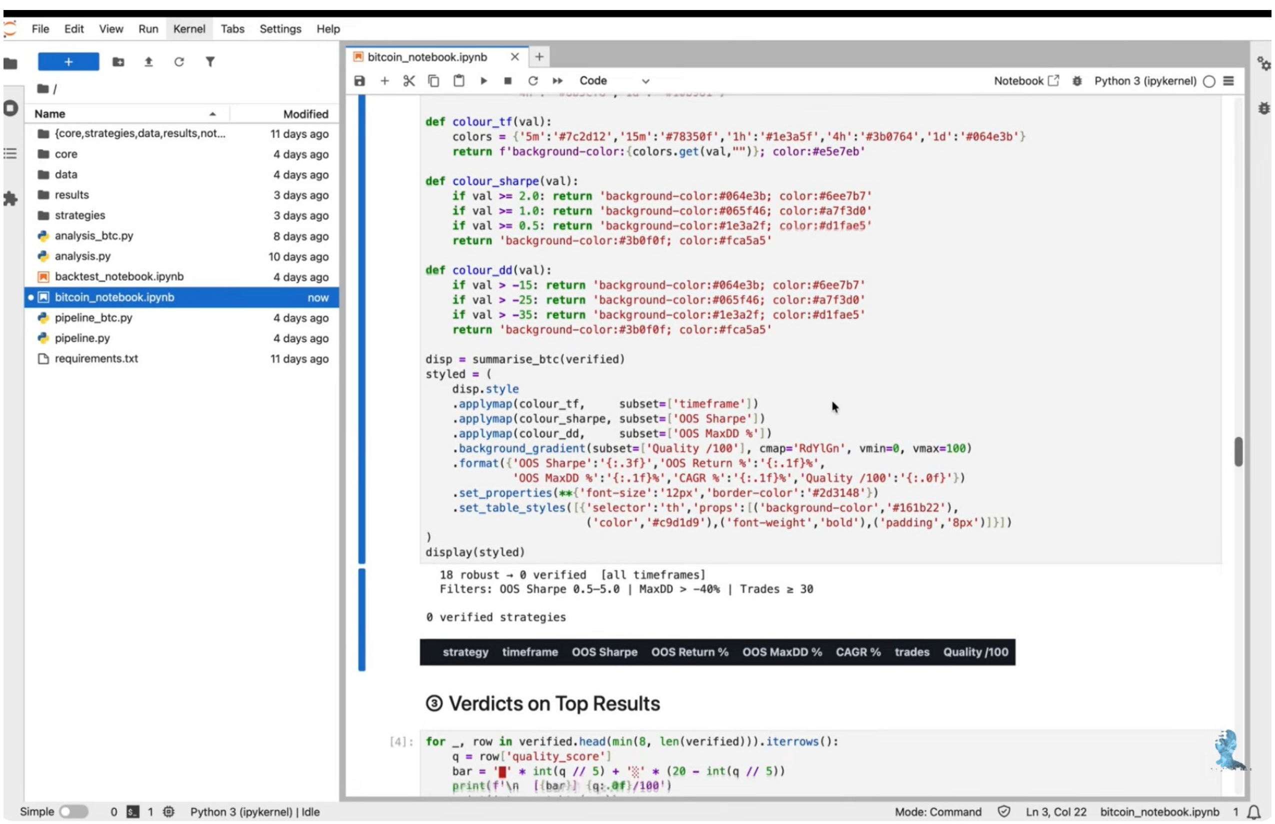
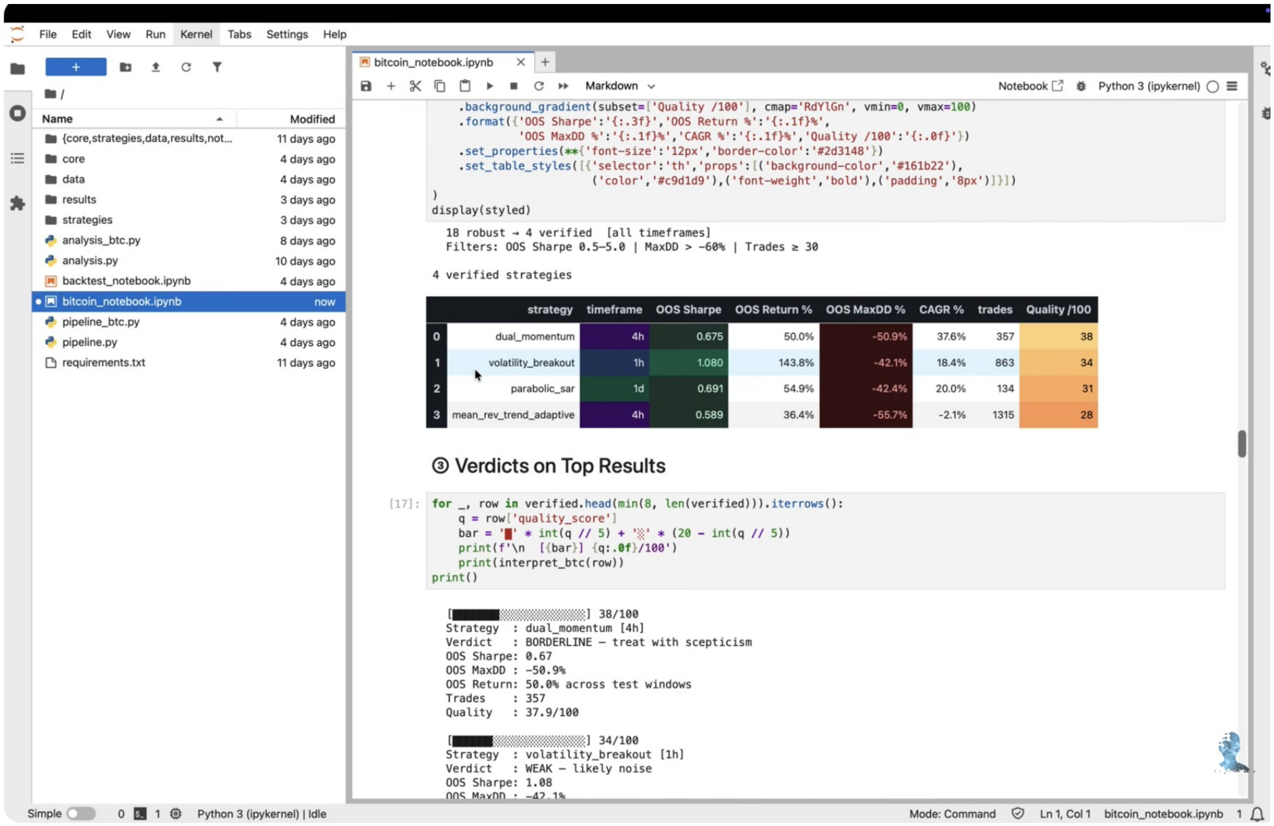


Figure 16. Video 2 screenshot 6 - Four verified at relaxed drawdown



13. Source Files Used

- Uploaded transcript: I Tested 100,000 Trading Strategies..txt
- Uploaded transcript: I Tested 350,000 Bitcoin Strategies. 4 Survived..txt
- User-provided screenshots from both YouTube strategy testing videos.
- Assistant analysis generated during this Matrix Trader 7.0 conversation.